

8 B

Apply Newton's second law to a geostationary satellite of mass m :

$$F = \frac{GM_E m}{r^2} = mr\omega^2 = mr\left(\frac{2\pi}{T}\right)^2$$

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$$\frac{GM_E}{r^3} = \frac{4\pi^2}{T^2} = \frac{4\pi^2}{(24 \text{ hours})^2}$$

Notice that m cancels out, so the geostationary orbit is independent of the mass of the satellite.

But the period and radius of orbit have unique values.